

Mary sneha Jeyakumar

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[LinkedIn](#) | [GitHub](#) | [Portfolio link](#)

Executive Summary

Data Science undergraduate with hands-on experience developing machine learning and AI-driven solutions, from data processing and predictive modeling to API development and deployment preparation. Skilled in explainable AI, natural language processing, model evaluation, and REST API development through academic and personal projects. A proactive learner with strong analytical and problem-solving abilities, eager to apply data science and AI skills to real-world challenges while continuously expanding knowledge in modern AI technologies and MLOps.

Technical Skills

- **Languages:** Python, SQL, TypeScript, JavaScript
- **Machine Learning & Data Science:** Pandas, NumPy, Scikit-learn, XGBoost, CatBoost, SHAP, EDA, feature engineering, model evaluation
- **AI & NLP:** OpenAI API, prompt engineering, text extraction, semantic matching
- **Backend & APIs:** FastAPI, REST APIs, Next.js API Routes, Pydantic, JWT, SQLAlchemy
- **Databases & Platforms:** PostgreSQL, Supabase, SQLite
- **Frontend & Tools:** Next.js, React, Tailwind CSS, Git, GitHub, Jupyter Notebook, Swagger/OpenAPI

Softskills

- Communication & Teamwork
- Time Management
- Analytical thinking and problem-solving
- Responsible and ethical AI awareness
- Adaptability
- Multilingual Abilities (English , Tamil and Sinhala — Basic)

Education

BSc (Hons) in Information Technology Specialization
in Data Science Sri Lanka Institute of Information Technology
Currently in Year 3

2024 – 2028 (Expected)

Projects :

AI Resume Analyzer

[Github](#)

- Built an end-to-end AI/NLP resume analysis platform using Next.js, Supabase, and OpenAI API to compare uploaded resumes against target job descriptions.
- Implemented PDF/DOCX resume text extraction using pdf-parse and mammoth, with Zod-based validation for structured AI report generation.
- Developed secure full-stack features including Supabase Auth, protected routes, PostgreSQL report storage, dashboard analytics, report history, delete functionality, and export/print-ready reports.

Fraud Detection ML System

[Github](#)

- Developed a machine learning fraud detection pipeline using Python, Scikit-learn, XGBoost, CatBoost, SHAP, and imbalanced classification techniques.
- Performed EDA, preprocessing, feature engineering, model comparison, threshold tuning, and explainability using fraud-focused metrics instead of relying on accuracy.
- Created a reusable prediction function to return fraud probability, risk level, and decision for new transactions.

SugarSense: Explainable ML-Based Diabetes Risk Estimation API

[Github](#)

- Trained and evaluated classification models for diabetes-related risk estimation, selecting Random Forest based on F1-score, recall, and ROC-AUC performance.
- Implemented isotonic probability calibration and SHAP explainability to provide calibrated risk scores and feature-level model explanations.
- Integrated the calibrated ML model into a FastAPI backend with JWT-secured endpoints, SQLite/SQLAlchemy history storage, analytics APIs, and PDF report generation.

Certificates

Full Stack Web Development — Uki | December 2025 | [[Certificate](#)]

- Practical training in React, Node.js, Express.js, MongoDB, REST APIs, TypeScript, Git and GitHub.
- Final Project: MERN Stack Pharmacy Portal